

ParkWatch Cloudflare Deployment Guide

Beginner steps for sharing ParkWatch globally for free using Cloudflare Quick Tunnel. This path does not need a paid domain.

Generated for the local project at `/home/kabeerkhosla/Desktop/flipkart/parkwatch`.

What This Gives You

- A public HTTPS link like `https://something.trycloudflare.com/dashboard`.
- Anyone on the internet can open the dashboard while your computer is on.
- No router setup, no port forwarding, and no domain purchase.
- The link is temporary and usually changes when you restart the tunnel.

Important: Without buying or owning a domain, Cloudflare Quick Tunnel is best for demos. It is not a permanent production URL. If the laptop sleeps, shuts down, or loses internet, the link stops working.

Before You Start

- [] Docker Desktop or Docker Engine is running.
- [] You are in the ParkWatch folder.
- [] The ParkWatch stack starts successfully with Docker Compose.
- [] The backend health check returns `{"status": "ok", "data_ready": true}`.

Step 1: Open The Project Folder

Open a terminal and run:

```
cd /home/kabeerkhosla/Desktop/flipkart/parkwatch
```

Step 2: Start ParkWatch

Run ParkWatch in the background:

```
docker compose up -d --build
```

Check that all containers are running:

```
docker compose ps
```

You should see:

- `parkwatch-backend` as healthy.
- `parkwatch-frontend` as healthy.

- `parkwatch-nginx` as running.

Step 3: Test Locally First

Open this in your browser:

```
http://localhost:3000/dashboard
```

Also check the backend:

```
curl http://localhost:8001/api/health
```

Expected output:

```
{"status":"ok","data_ready":true}
```

Step 4: Install cloudflared

If `cloudflared` is not installed, install it on Linux with:

```
wget https://github.com/cloudflare/cloudflared/releases/latest/download/cloudflared-linux-amd64.deb  
sudo dpkg -i cloudflared-linux-amd64.deb
```

Check installation:

```
cloudflared --version
```

Step 5: Create The Public Cloudflare Link

In a new terminal, run:

```
cloudflared tunnel --url http://localhost:3000
```

Cloudflare will print a URL that looks like this:

```
https://example-random-name.trycloudflare.com
```

Open the dashboard by adding `/dashboard` :

```
https://example-random-name.trycloudflare.com/dashboard
```

Done: That `trycloudflare.com` link is your global public ParkWatch link. Share it with anyone while the tunnel terminal stays open.

Step 6: Keep It Online

- Keep Docker running.
- Keep the `cloudflared tunnel` terminal open.
- Keep the laptop awake.
- Keep your internet connection stable.

Daily Start Commands

When you want to share ParkWatch again later:

```
cd /home/kabeerkhosla/Desktop/flipkart/parkwatch
docker compose up -d --build
cloudflared tunnel --url http://localhost:3000
```

Then copy the new `trycloudflare.com` link and open `/dashboard`.

Stop Everything

Stop the Cloudflare tunnel with `Ctrl+C` in the tunnel terminal.

Stop ParkWatch with:

```
cd /home/kabeerkhosla/Desktop/flipkart/parkwatch
docker compose down
```

Troubleshooting

The Cloudflare URL opens but dashboard data does not load

```
docker compose ps
curl http://localhost:8001/api/health
```

If the backend is not healthy, restart:

```
docker compose down
docker compose up -d --build
```

Port 8000 is busy

This project currently maps the backend to `8001` on your laptop because another container was using host networking. Use:

```
curl http://localhost:8001/api/health
```

Port 3000 is busy

Find what is using it:

```
ss -ltnp 'sport = :3000'
```

Either stop that app or change the frontend port in `docker-compose.yml`.

Cloudflare link changed

This is normal for Quick Tunnel. Run the tunnel command again and use the new URL.

You want a permanent Cloudflare URL

A permanent Cloudflare Tunnel hostname requires a domain managed in Cloudflare DNS. That can still be low cost, but it is no longer the no-domain free path.

Useful Commands

```
docker compose ps
docker compose logs -f
docker compose logs -f backend
docker compose logs -f frontend
docker compose down
cloudflared tunnel --url http://localhost:3000
```

Official References

- Cloudflare Quick Tunnels: <https://developers.cloudflare.com/cloudflare-one/networks/connectors/cloudflare-tunnel/do-more-with-tunnels/trycloudflare/>
- Cloudflared downloads: <https://developers.cloudflare.com/cloudflare-one/networks/connectors/cloudflare-tunnel/downloads/>